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TEAM ENROLLMENT CONFIRMATION

FitBuddy – AI Fitness Plan Generator using Gemini Models

Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation - FitBuddy – AI Fitness Plan Generator using Gemini Mode
Team Members	Divya.D, Deepika.M, Dharani.T, Divya shree.R, DivyaDharshini.V

Project Type: Generative AI / Fitness & Wellness Application
Primary Model: Google Gemini models
Suggested Stack: Python, Flask, HTML, CSS, JavaScript, Gemini API

This document presents the project overview, scenarios, architecture, prerequisites, workflow, milestones, team responsibilities, testing plan and conclusion for FitBuddy.

Project Documentation

Project Description

FitBuddy is an AI-powered fitness plan generator designed to create personalized and easy-to-follow fitness guidance from user-provided goals and preferences. The application uses Gemini models to transform inputs such as fitness goal, activity level, preferred workout style, available time and equipment into a structured plan.

The system is intended as a planning and educational assistant. It can generate workout schedules, exercise suggestions, recovery guidance and motivational tips through a simple web interface.

Key Features

- **AI Fitness Plan Generator:** Produces a structured plan based on user requirements.
- **Goal-Based Personalization:** Supports general fitness, strength, endurance and flexibility goals.
- **Workout Preferences:** Uses activity level, time availability, equipment and preferred workout style.
- **Gemini Integration:** Sends structured prompts to a Gemini model and displays the response.

- **Simple User Interface:** Collects inputs through forms and presents the generated plan clearly.
- **Regeneration:** Allows users to modify inputs and request another plan.

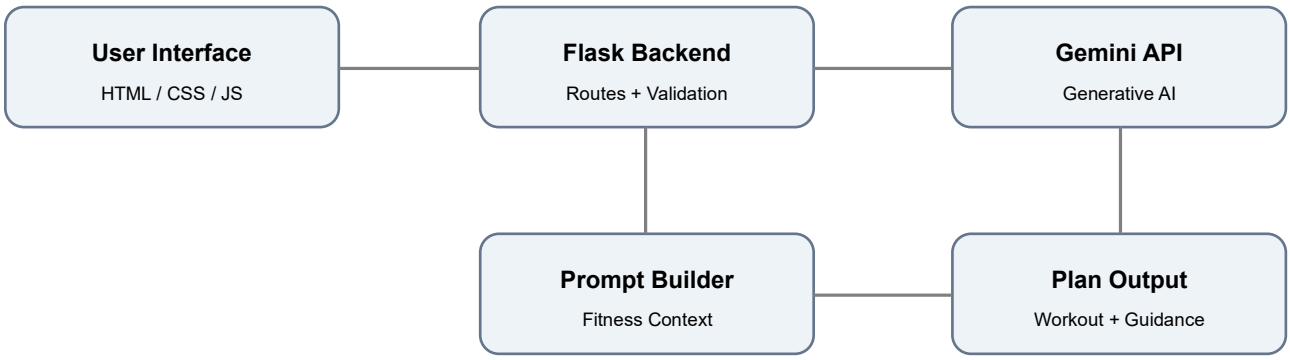
Scenarios

Scenario 1: A beginner selects general fitness, low activity, 30 minutes and no equipment. FitBuddy generates a simple weekly routine with warm-up, exercises, cool-down and rest guidance.

Scenario 2: A user interested in strength selects moderate activity and home equipment. FitBuddy generates a structured plan with exercise categories and recovery guidance.

Scenario 3: A user with limited time selects a 20-minute workout preference. FitBuddy generates a concise routine that fits the selected time.

Technical Architecture



Architecture flow: User → Flask backend → prompt builder → Gemini API → generated fitness plan → user interface.

Description

FitBuddy follows a modular web-application architecture. The frontend collects user preferences and sends them to the Flask backend. The backend validates the input, builds a structured prompt and communicates with the Gemini API. The response is processed and displayed as a readable fitness plan. API credentials should be stored securely in environment variables.

Note: If a database is added for user profiles, saved plans or progress history, an ER diagram and database description should be included.

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Prerequisites

- Python programming fundamentals
- Flask framework knowledge
- HTML, CSS and basic JavaScript
- Gemini API familiarity and API-key handling
- REST-style request/response flow
- Git and GitHub basics
- VS Code or another development environment

Project Workflow

Milestone 1: Model Selection and Architecture

- **Activity 1:** Define requirements, target users and fitness-plan inputs.
- **Activity 2:** Configure Gemini API access and select a suitable Gemini model.
- **Activity 3:** Design interactions between frontend, Flask backend and Gemini API.
- **Activity 4:** Set up Python environment and required libraries.

Milestone 2: Core Functionalities Development

- **Activity 1:** Develop fitness-plan generation and structured prompt templates.
- **Activity 2:** Implement Flask routes for user inputs and AI responses.
- **Activity 3:** Add input validation and safe error handling.

Milestone 3: Application Development

- **Activity 1:** Connect frontend forms with the Flask backend.
- **Activity 2:** Integrate Gemini responses and format the generated plan.
- **Activity 3:** Add reset/regeneration functionality.

Milestone 4: Frontend Development

- **Activity 1:** Design a responsive interface using HTML and CSS.
- **Activity 2:** Add JavaScript interactions and user feedback.
- **Activity 3:** Create clear result sections for workouts and guidance.

Milestone 5: Testing and Deployment

- **Activity 1:** Test different goals, activity levels, durations and equipment.
- **Activity 2:** Check API errors, empty inputs and unexpected responses.
- **Activity 3:** Prepare the application for local or approved public deployment.

Milestone 6: Conclusion

- **Activity 1:** Review the final system, document results, limitations and future enhancements.

Milestone 1: Model Selection and Architecture

This milestone focuses on defining FitBuddy requirements, selecting a Gemini model and creating the technical structure needed for reliable AI-generated fitness plans.

Activity 1.1: Define User Inputs

Input	Purpose
Fitness Goal	Defines the main objective.
Activity Level	Helps adapt suggested difficulty.
Workout Duration	Keeps the plan within available time.
Equipment	Determines suitable exercise options.
Preferred Workout Type	Personalizes the routine style.
Days per Week	Controls the weekly schedule.

Activity 1.2: Gemini API Configuration

- Create/use an approved Gemini API key through the relevant Google AI development platform.
- Store the key in an environment variable such as GEMINI_API_KEY.
- Never commit the secret key to a public Git repository.
- Configure the selected Gemini model and generation settings in the backend.

Activity 1.3: Prompt Design

The prompt should include goal, activity level, duration, equipment and schedule. The expected output can be requested as weekly overview, daily routine, warm-up, main workout, cool-down, recovery notes and safety reminders.

Milestone 2: Core Functionalities Development

- **Fitness Plan Generator:** Accepts validated preferences and requests a personalized plan from Gemini.
- **Prompt Builder:** Converts form data into a clear and consistent AI prompt.
- **Response Formatter:** Organizes the model response into readable sections.
- **Error Handling:** Shows a useful message when input is missing or the AI request fails.
- **Regeneration:** Lets users change preferences and generate a revised plan.

Milestone 3: Application Development

Component	Responsibility
Home page	Display the main FitBuddy input form.
/generate	Receive preferences, call Gemini and return the generated plan.
Gemini Service	Handle API communication and generation configuration.
Validation Layer	Check required fields and acceptable input values.
Result Template	Display the generated plan in a structured format.

Expected output: A personalized weekly plan with exercise suggestions, time structure, rest/recovery guidance and general safety reminders.

Milestone 4: Frontend Development

- Create a simple FitBuddy landing/input page.
- Use clear labels and selectable options for goals, activity level, duration, equipment and days.
- Add a Generate Plan button and a loading/status message.
- Display results using readable sections or cards.
- Provide Reset / Regenerate controls.

Milestone 5: Testing and Deployment

Test Area	Example Test	Expected Result
Required fields	Submit with empty goal	Validation message appears.
Normal generation	Use valid preferences	Structured plan is returned.
Short duration	Select 15–20 minutes	Plan respects selected duration.
Equipment	Select no equipment	Plan avoids requiring equipment.
API failure	Simulate unavailable API	Friendly error message.
Mobile layout	Open on phone width	Inputs/results remain readable.

For deployment, keep API secrets outside source code, configure environment variables, install dependencies from the requirements file and verify the application using test inputs.

Team Members and Work Allocation

S.No.	Team Member	Assigned Work
1	Divya.D	Project Lead & Backend Setup — requirements, Flask structure, API configuration, backend
2	Deepika.M	Frontend Development — HTML/CSS interface, form design, responsive layout, result displ
3	Dharani.T	Gemini AI Integration — prompt design, Gemini model integration, response handling and A
4	Divya shree.R	Fitness Plan Logic & Validation — input fields, workout-plan structure, validation rules, test
5	DivyaDharshini.V	Testing, Documentation & Deployment — functional testing, error cases, documentation, sc

Responsibility Flow

Divya.D	→	Deepika.M	→	Dharani.T	→	Divya shree.R	→	DivyaDharshini.V
Backend		Frontend		Gemini AI		Plan Logic		Testing & Docs

Milestone 6: Conclusion

FitBuddy demonstrates how Gemini generative AI can be integrated into a web application to produce personalized fitness-plan content from structured user preferences. The modular design separates the interface, backend processing, prompt construction and AI integration, making the application easier to test and extend.

Future Scope

- User accounts and saved plans
- Progress tracking and history
- Exercise illustrations or media
- Multilingual support
- Calendar scheduling
- Optional integration with approved activity data sources

Important Safety Note

FitBuddy should present AI-generated plans as general wellness information rather than medical advice. Users should consider their own health circumstances and consult a qualified professional when appropriate, especially before starting a new exercise program.

Project Deliverables

- Working FitBuddy web application
- Gemini API integration and prompt configuration
- Frontend input and result pages
- Testing checklist and results
- Source code with secure environment-variable setup
- Final project documentation

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